Foundations · Six parts, one workflow

Workflow Design & Diagnosis

Which steps AI should own, and what to do when it goes wrong.

Made by Taha Shabbir

Based on The AI Agent Factory by Panaversity · Foundations

The output is not the verdict

Two teams, one tool. One mapped the work step by step: review time halved, quality held. The other let AI approve unsupervised: an obligation slipped through and the tool was pulled.

Which step
A step is decided in advance. A session is judged afterwards.

What it costs
In the bad case, not on average.

Who answers
Accountability never moves onto the machine.

When it started
Timing tells you where to look first.

Stage	The question that decides it
Design	What are the actual requirements, and which numbers must be computed?
Map	For this step: can it be undone, what does an error cost, who answers?
Build	Does this still fit in a prompt-and-iterate loop, or has it outgrown it?
Diagnose	When did the symptom first appear?
Persist	Is this fix a rule, a reference, or a procedure?
Communicate	What does it do, and where exactly is the human gate?

Parts 1 & 3 · Design first, then build

Requirements first

Extract every requirement into a table, then attack it: which are ambiguous? If you cannot write a check for it, it is not decidable yet.

Computed, never written

If a result matters, compute it with code execution. A calculation asked for in prose is why totals come back subtly wrong.

The window test

Could I put everything this step needs into the window? If not, the step is mine. Passing makes a step eligible for AI, never owned by AI.

Part 3 · Build in a loop that can fail quietly

Ideate, prototype, feedback, refine, inside a Project. Cycle 1: build the dashboard. Cycle 2: totals are wrong, so move them to code execution. Cycle 3: colour and print, solved by describing better.

Escalate when three departments open it every Monday: dependency means uptime, access, someone to call. That is a correct answer, not an admission.

Part 2 · The delegation map

1 · Eligibility
May this data enter this tool? Can everything the step needs be supplied? A no ends the question.

2 · Ownership
Reversibility, stakes, accountability. Name the one that decided it. Where they disagree, the strictest wins.

3 · Implementation
Skill, knowledge, code execution, instruction or gate. Human steps need a named person, moment and check.

Contract review step	Owner	Criterion
Extract clauses	AI	Reversible, low stakes
Flag playbook departures	AI	Errors surface at redline
Draft the redline	Collaborative	High stakes, human judges
Compute penalty exposure	AI	Checked at approval gate
Approve or reject	Human	Decision somebody answers for
Sign and send	Human	Irreversible, binding

Three ways a map goes wrong

Halo delegation: handed over because the last step went well.

The unstaffed gate: review becomes a glance, then a click.

Mapping the tool, not the work: the workflow grows a step shaped like the Skill.

Part 4 · Diagnose by timing

Before changing anything, ask one question: when did the symptom first appear?

Symptom appeared	First hypothesis	Cheap confirmation	Fix
Wrong from the first answer	Under-specification	Is the missing thing in the prompt?	Add what it left out
Started fine, then degraded	Context overload	Restate it in a fresh session	Restart or summarise
One repeatable error type	Wrong feature or model	Run once with the right feature or tier	Change feature or tier
Used to work, now does not	Stale configuration	Check the instruction's date	Maintain the configuration

The diagnostic sequence, cheapest first

1 Re-read the prompt: role, context, task, constraints, format.

2 Check the conversation length.

3 Check the feature and the model.

4 Check the configuration.

5 Only then ask whether the task fits.

Parts 5 & 6 · Persist, measure, communicate

Persist the fix

Turn a reaction into an instruction and name the lever: prompt, instruction or knowledge. A repeated correction becomes a rule (standing instruction), a reference (knowledge base) or a procedure (Skill).

Measure the before

Run the old way once: time, revision rounds, manual steps. Then run old and new in parallel for two or three cycles. Optimise for time, consistency, accuracy or variance.

Communicate

Say what it does, then name the gate, in the detail each audience needs: technical colleague, executive, risk function. Never write "fully automated".

The one-page map

Header with a named owner and review dates, steps (owner, criterion, carried by), baseline, eligibility, gates (who, when, checking what) and what to do if it goes wrong.

The output is not the verdict. Judge the step by what it costs and who answers for it. Judge a failure by when it started.

Practice 1 · Scenario

Hira leads a finance team. Each week AI pulls invoices from the ERP export, summarises exceptions and drafts the payment batch. A clerk used to approve every batch. Because the summaries have been accurate for three months, the team now lets AI release any batch under $5,000 on its own. Last week a duplicate invoice was paid twice, and the money cannot be recalled.

Which is the first step where the team's map departs from the method?

A. Pulling invoices from the ERP export, because retrieval from a finance system must always be done by a person.

B. Summarising exceptions, because a Skill that has been accurate for three months must be re-tested before every run.

C. Releasing payments, because it is irreversible and is the decision somebody answers for. Accurate summaries say nothing about this step.

D. Drafting the batch, because AI should never prepare any document that involves money.

Practice 2 · Scenario with table

Sana's monthly sales report was correct for five months, then it began including cancelled orders. Her notes:

#	Question	What she recorded
1	When did it start?	Correct for five months. Wrong for the first time this month.
2	First hypothesis	Under-specification, so she rewrote the whole prompt.
3	Confirmation	The new prompt gave the same error on two runs.
4	Next hypothesis	Wrong model, so she moved to the strongest tier. Same error.
5	Conclusion	"AI is not a fit for this report." Back to manual.

Where does her reasoning first depart from the method?

A. Step 2: "used to work, now does not" points first to stale configuration. She should have checked the instruction's date.

B. Step 1: she should have run a baseline before noticing any symptom.

C. Step 4: trying a stronger model tier is never a valid hypothesis.

D. Step 5: reports are high-stakes, so AI should never have owned this step.